

Message typing UI

The keyboard UI will be designed to input alphanumeric characters and symbols for typing and sending messages. This process is similar to the dial pad but with more touch keys on the display. A special key will be added to switch between alphabets, symbols, special characters, and to toggle capitalisation. Fig. 6 shows the UI for the message and keypad.

The basic UI is now ready for functions such as sending messages, dialling numbers, and checking incoming calls. The remaining UI will be covered in the next part. Now, let us turn our attention to integrating the GSM SIM card.

GSM integration

The GSM module operates through the serial peripheral and connects to the IndusBoard coin using the serial port. The flexible I/O of IndusBoard allows the addition of many serial peripheral devices. Defining pins in the code will enable them to function as serial pins. Most GSM modules communicate with the IndusBoard coin using AT commands, and most AT commands are standard across modules. Functions such as adjusting volume, making calls, checking signal strength, network provider status, message storage, and more can be handled via AT commands. Libraries simplify access to these functions, and for this device, the GSMSIM library is utilised. The code can be uploaded to make calls and check signal strength, and functions can be tested as required. Fig. 7 shows the code snippet for testing GSM and making calls or sending messages. Fig. 8 shows the 4G GSM module.

EFY note. Any 3G, 2G, or 4G GSM module can be used for the SIM card. To enable 4G

```

1 void setup() {
2   // Initialize serial communication at 9600 baud rate
3   Serial.begin(9600);
4   Serial.println("SIM800L Simulator Ready...");
5 }
6
7 void loop() {
8   // Check if data is available on the Serial input
9   if (Serial.available()) {
10    String command = Serial.readStringUntil('\n');
11    command.trim(); // Remove any trailing whitespace or newline characters
12
13    if (command == "AT") {
14      Serial.println("OK");
15    }
16    else if (command == "AT+CSQ") { // Signal Quality
17      Serial.println("+CSQ: 20,0");
18    }
19    else if (command == "AT+COPS?") { // Check Operator
20      Serial.println("+COPS: 0,0,\"VodafoneIdea\",0");
21    }
22    else if (command == "AT+CLCC") { // List Current Calls
23      Serial.println("NO CARRIER");
24    }
25    else if (command == "AT+CCLK?") { // Query Local Time
26      Serial.println("+CCLK: \"23/08/09,12:34:56+08\"");
27    }
28    else if (command == "AT+CLVL?") { // Query Volume Level
29      Serial.println("+CLVL: 3");
30    }
31    else if (command == "AT+CREG?") { // Network Registration Status
32      Serial.println("+CREG: 0,1");
33    }
34    else if (command == "AT+CGMI") { // Manufacturer Identification
35      Serial.println("SIMCOM_Ltd");
36    }
37    else if (command == "AT+CGMM") { // Model Identification
38      Serial.println("SIM800L");
39    }
40    else if (command == "AT+CGSN") { // Serial Number (IMEI)
41      Serial.println("123456789012345");
42    }
43    else {
44      // If an unknown command is received, respond with an error
45      Serial.println("ERROR");
46    }
47  }
48 }

```

Fig. 7: Code snippet for testing GSM, making calls, and messages

functionality, the A7670C module is recommended. Alternatively, a 2G module, such as the SIM800L, can be used if 4G support is not required.

Circuit diagram

The circuit diagram for the smallest phone is shown in Fig. 9. It is built around IndusBoard coin, a GSM module, a round touchscreen display, a round LCD, and a few other components.



Fig. 8: 4G GSM module